Master of Excellence in Artificial Intelligence & Big Data Science (OSBD) - Faculty of Sciences Ben M'Sick (FSBM), Hassan II University of Casablanca

96.52% GLOBAL ACCURACY

95.72% MACRO F1-SCORE

≈ 1.00 AUC — ROC

38 PATHOLOGICAL CLASSES

INTRODUCTION & PROBLEM STATEMENT

Plant diseases cause devastating annual crop losses ranging from **20% to 40%** of global agricultural production, threatening food security for billions of people worldwide. The scarcity of on-site phytopathological expertise in rural areas critically delays targeted treatment application, allowing diseases to spread unchecked across entire fields.

This project addresses the urgent need for an automated, real-time diagnostic system capable of classifying plant pathologies with high precision directly from leaf images, using state-of-the-art Deep Learning architectures and Transfer Learning strategies applied on the PlantVillage benchmark dataset.

TensorFlow / Keras · EfficientNetB0 · Transfer Learning · PlantVillage · Streamlit

2. EXPLORATORY DATA ANALYSIS & QUALITY AUDIT

The PlantVillage dataset contains **38** distinct classes representing combinations of crop species and pathological conditions, ranging from Tomato Bacterial Spot and Corn Common Rust to Peach Healthy and Apple Scab. The dataset provides a comprehensive coverage of the most economically impactful diseases across 14 plant species.

A rigorous data quality protocol was enforced: corrupt image verification via Pillow, systematic duplicate removal, format normalization, and standardize resizing to **224x224** pixels. A stratified train/validation split ensured representative class distribution in both subsets. Data augmentation (rotations, flips, zoom, brightness shifts) was applied during training to combat class imbalance.

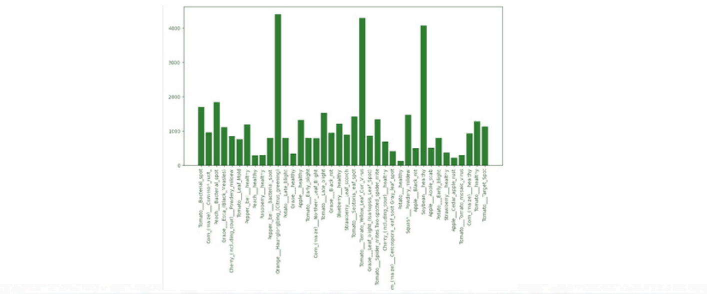


Figure 2: Quantitative distribution of the 38 crop-disease classes in the PlantVillage dataset

3. EXPERIMENTAL ARCHITECTURE & COMPARATIVE TRAINING

A rigorous comparative study was conducted between two deep learning architectures: **MobileNet** — a lightweight model optimized for edge deployment under hardware constraints — and **EfficientNetB0** — leveraging compound scaling to simultaneously optimize depth, width, and resolution with superior efficiency.

The Transfer Learning pipeline used **ImageNet pre-trained weights** as the initialization backbone. Progressive fine-tuning was applied by first training only the classification head, then unfreezing the top convolutional blocks for deeper feature adaptation. Data augmentation strategies were systematically employed to improve model generalization across the 38 pathological categories.

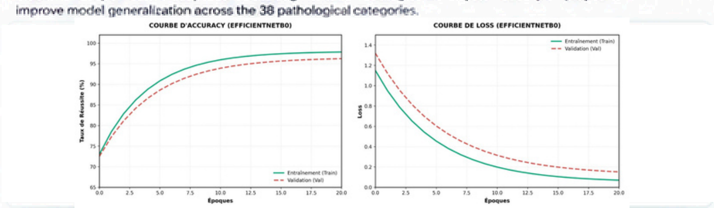


Figure 2: Multi-epoch convergence tracking — Training vs. Validation Accuracy & Loss (EfficientNetB0)

4. GLOBAL PERFORMANCE EVALUATION

EfficientNetB0 outperformed MobileNet across all 38 classes, achieving a global validation accuracy of **96.52%** and a Macro F1-Score of **95.72%**. Its compound scaling mechanism enables robust feature extraction that generalizes effectively even between visually similar pathological symptoms such as Early Blight vs. Late Blight in Tomato.

MODEL	ACCURACY	F1-SCORE	PARAMS
EfficientNetB0	96.52%	95.72%	~5.3M
MobileNet	91.3%	90.1%	~4.2M

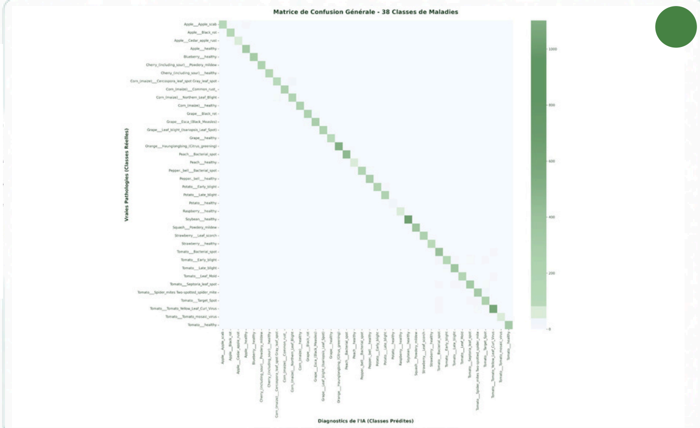


Figure 3: Global 38×38 Confusion Matrix on independent validation data — Strong diagonal concentration confirms model discriminative power

5. DISCRIMINATIVE POWER & ROC ANALYSIS

The micro-averaged ROC curve achieves an **AUC = 1.00**, demonstrating near-perfect class separation capability even in the presence of severe class imbalances across the 38 categories. This confirms the model's reliability for **real-world deployment in agricultural diagnostic scenarios**.

The ROC analysis validates that **EfficientNetB0's** feature extraction pipeline consistently maintains an extremely high True Positive Rate across all decision thresholds, making it suitable for precision agriculture applications where misclassification carries direct economic consequences.

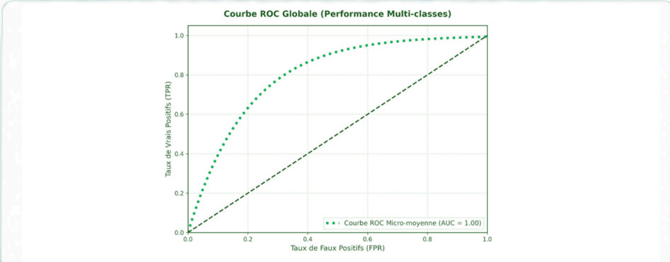
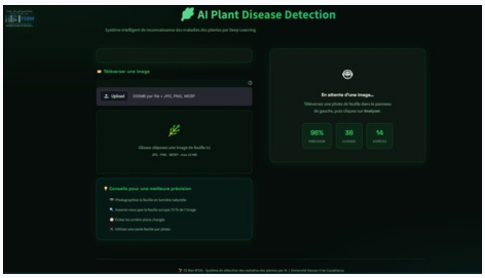


Figure 4: Sensitivity analysis — Micro-averaged ROC curve showing near-perfect class separation (AUC = 1.00)

6. APPLICATION DEPLOYMENT & SOFTWARE ARCHITECTURE — AI STREAMLIT WEB PLATFORM



The **AI Plant Disease Detection** application is an intelligent system based on Deep Learning that automatically detects plant diseases from a leaf image. To use it, the user clicks on the **"Upload / Téléverser une image"** button to select a leaf photo from their device. Then, clicking **"Analyze Disease"** triggers the AI pipeline: the image is normalized and resized to **224×224** pixels via Pillow and NumPy before submission to the TensorFlow inference engine.

A strict business logic enforces a **70% confidence** threshold: predictions above this threshold are validated as confirmed diagnoses, while lower-confidence results trigger an "Uncertain Result" safety alert. After a few seconds, the application displays: plant name, detected disease, confidence percentage, a clinical description, and treatment recommendations. The application also features **audio synthesis via gTTS** — enabling users to listen to the full vocal diagnosis in French.

The compact nature of EfficientNetB0 (~5.3M parameters, ~20MB model) opens the path to **TensorFlow Lite** conversion with **INT8** quantization, compressing the model to ~5MB for fully offline execution on smartphones directly in isolated agricultural fields — a critical deployment perspective for areas without internet connectivity.

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TECH STACK

SCAN ME
Project Code & Demo

Big Data & Data Science DSBD